Data Cleaning & Reporting Automation

Project Overview

Data Cleaning & Reporting Automation is a project focused on automating the process of cleaning raw data and generating reports. The objective is to improve data quality, reduce manual effort, and create efficient reporting workflows.

Project Objective

To automate data cleaning and reporting workflows using Python, Excel, or Power BI while ensuring data accuracy and reporting efficiency.

Key Features

- Automated Data Cleaning
- Missing Value Handling
- Duplicate Record Removal
- Inconsistent Data Correction
- Automated Report Generation
- Visual Summaries and Dashboards
- Data Quality Monitoring

Tools Used

- Python
- Pandas
- NumPy
- OpenPyXL
- Microsoft Excel
- Power BI

Dataset Requirements

The dataset may include:

- Customer Data
- Sales Data
- Employee Data

- Product Data
- Financial Records

Example Fields:

- Customer ID
- Name
- Email
- Date
- Sales Amount
- Region
- Category

Data Cleaning Process

Step 1: Import Dataset

Read data from:

- CSV Files
- Excel Files
- Databases

Step 2: Handle Missing Values

Techniques:

- Remove Null Records
- Fill Missing Values
- Replace Missing Data with Mean/Median

Step 3: Remove Duplicates

Identify duplicate records and remove them.

Step 4: Standardize Data

Examples:

- Consistent Date Formats
- Proper Capitalization
- Standard Category Names

Step 5: Data Validation

Validate:

- Data Types
- Numeric Ranges
- Invalid Entries

Python Automation Code

```
import pandas as pd

df = pd.read_csv('sales_data.csv')

df.drop_duplicates(inplace=True)

df.fillna(method='ffill', inplace=True)

df.to_excel('cleaned_data.xlsx', index=False)

print("Data Cleaning Completed")
```

Automated Reporting

Generate reports automatically using:

Excel Reports

- Pivot Tables
- Charts
- Summary Reports

Power BI Reports

- KPI Cards
- Interactive Dashboards
- Trend Analysis

Python Reports

- Automated Excel Reports
- PDF Reports
- Email Reports

Key Performance Indicators (KPIs)

- Total Records Processed
- Missing Values Removed
- Duplicate Records Removed
- Data Accuracy Percentage
- Processing Time

Reporting Dashboard Components

KPI Cards

- Total Records
- Clean Records
- Errors Fixed

Charts

- Missing Values Trend
- Data Quality Score
- Records Processed by Category

Filters

- Date Range
- Category
- Region

Business Benefits

- Improved Data Quality
- Faster Reporting
- Reduced Manual Work
- Better Decision-Making
- Increased Productivity

Expected Outcome

By completing this project, you will gain experience in:

- Data Preprocessing
- Data Cleaning Techniques
- Workflow Automation
- Automated Reporting
- Dashboard Development
- Business Analytics

GitHub Repository Description

Data Cleaning & Reporting Automation project using Python, Excel, and Power BI to automate data preprocessing, handle missing and inconsistent data, and generate automated reports with visual summaries.

README Description

A data automation project focused on cleaning raw datasets, improving data quality, and generating automated reports and dashboards for efficient business reporting.

Repository Structure

```
data-cleaning-reporting-automation/
│
├── README.md
├── data_cleaning_report.docx
├── automation_script.py
├── dataset.csv
├── cleaned_data.xlsx
├── requirements.txt
└── screenshots/
```

Submission Link Example

https://github.com/shiprasahani89-max/data-cleaning-reporting-automation

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Resume Description

Developed a Data Cleaning & Reporting Automation solution using Python, Excel, and Power BI. Automated data preprocessing workflows, handled missing and duplicate records, generated automated reports, and improved reporting efficiency through data visualization and automation techniques.